Beam Elements

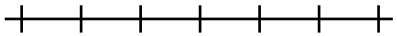


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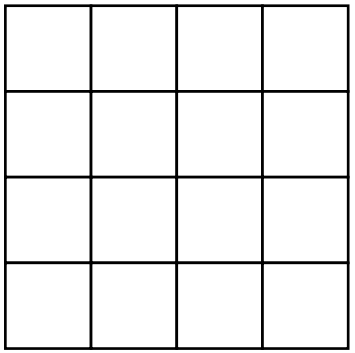
Finite Elements Method summary

Solve partial differential equations in *spatial* domains by spliting such domain in smaller subdomains (elements)

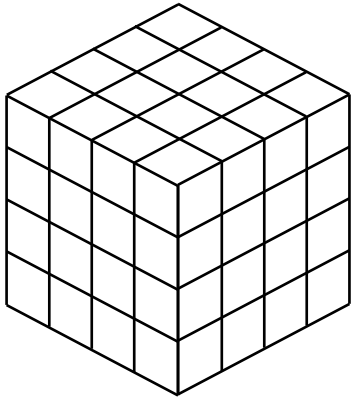
1D



2D



3D



Step I

Partial differential equation/s (*strong form*)

$$\Pi\big(u_i(\boldsymbol{x})\big) = 0$$

+

boundary conditions for $u_i(\boldsymbol{x})$

$u_i(\boldsymbol{x})$: Arguments (degrees of freedom)

Partial differential equation/s (*weak form*)

$$\sum_j \int_\Omega \Pi\big(u_i(\boldsymbol{x})\big)\delta u_j(\boldsymbol{x})\mathrm{d}\Omega = 0$$

+

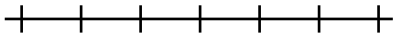
boundary conditions for $u_i(\boldsymbol{x})$, for all $\delta u_j(\boldsymbol{x})$ satisfying (null) boundary conditions

$\delta u_j(\boldsymbol{x})$: Test function

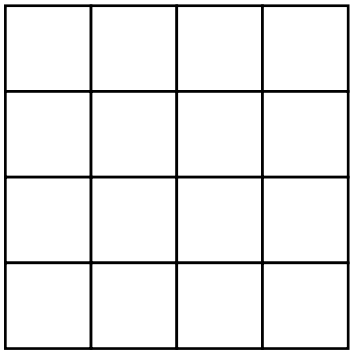
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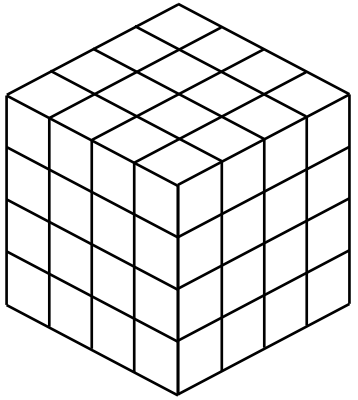
1D



2D



3D



Step II

Partial differential equation/s (*weak form*)

$$\sum_j \int_\Omega \Pi\big(u_i(\boldsymbol{x})\big)\delta u_j(\boldsymbol{x})\mathrm{d}\Omega = 0$$

+

boundary conditions for $u_i(\boldsymbol{x})$, for all $\delta u_j(\boldsymbol{x})$ satisfying (null) boundary conditions

$\delta u_j(\boldsymbol{x})$: Test function

Discretization

$$\sum_j \sum_e \int_{\Omega^{(e)}} \Pi'\left(\hat{u}_i^{(m)}\right)\delta\hat{u}_j^{(n)}\mathrm{d}\Omega = 0$$

- Domain split into elements $\Omega^{(e)}$
- Arguments are interpolated into each element from the values at discrete points (nodes):

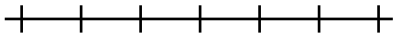
$$u_i(\boldsymbol{x}) = N_i^{(e,k)}(\boldsymbol{x})\hat{u}_i^{(e,k)}$$

$N_i^{(e,k)}(\boldsymbol{x})$: Interpolation function (shape function)

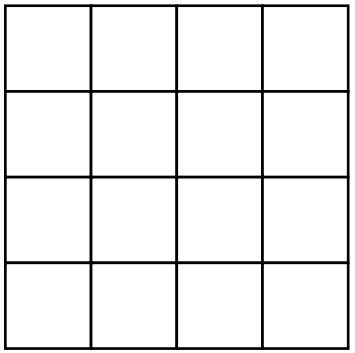
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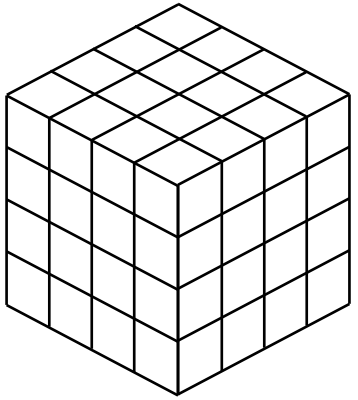
1D



2D



3D



Integration

$$\sum_j \sum_e \int_{\Omega^{(e)}} \Pi'\left(\hat{u}_i^{(m)}\right) \delta\hat{u}_j^{(n)} d\Omega = 0$$

Step III

$$\int_{\Omega^{(e)}} \Pi'\left(\hat{u}_i^{(m)}\right) \delta\hat{u}_j^{(n)} \mathrm{d}\Omega = \sum_l \sum_k \sum_i \delta\hat{u}_j^{(e,l)} \underbrace{\int_{\Omega^{(e)}} \overbrace{\Psi\left(N_j^{(e,l)}(\boldsymbol{x}), N_i^{(e,k)}(\boldsymbol{x})\right)}^{\text{Function of the shape functions}} \mathrm{d}\Omega}_{\text{Gauss quadrature (*)}} \hat{u}_i^{(e,k)}$$

(*) Change of variable: $\boldsymbol{x} \rightarrow \boldsymbol{\xi} \quad (-1 < \xi_i < 1)$

$$\frac{\partial}{\partial \boldsymbol{\xi}} = \boldsymbol{\mathcal{J}} \frac{\partial}{\partial \boldsymbol{x}}$$

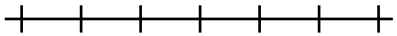
$\boldsymbol{\mathcal{J}}$: Jacobian matrix

$$\sum_a w_a |\boldsymbol{\mathcal{J}}(\boldsymbol{\xi}_a)| \Psi\left(N_j^{(e,l)}(\boldsymbol{\xi}_a), N_i^{(e,k)}(\boldsymbol{\xi}_a)\right)$$

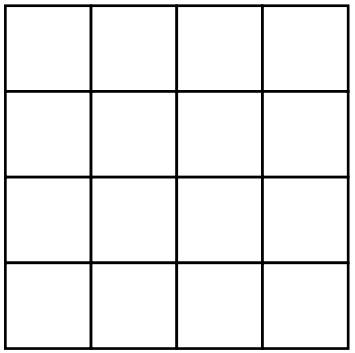
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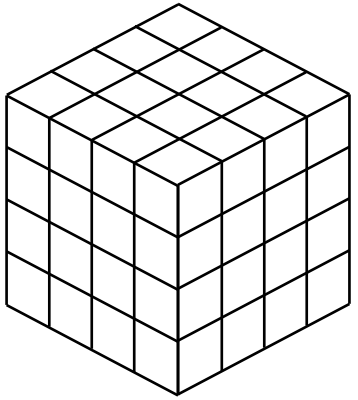
1D



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Matrix form

Let's define the coefficient:

$$K^{(e)}_{(j,l)(i,k)} = \sum_a w_a |\mathcal{J}(\xi_a)| \Psi\left(N_j^{(e,l)}(\xi_a), N_i^{(e,k)}(\xi_a)\right)$$

Step IV

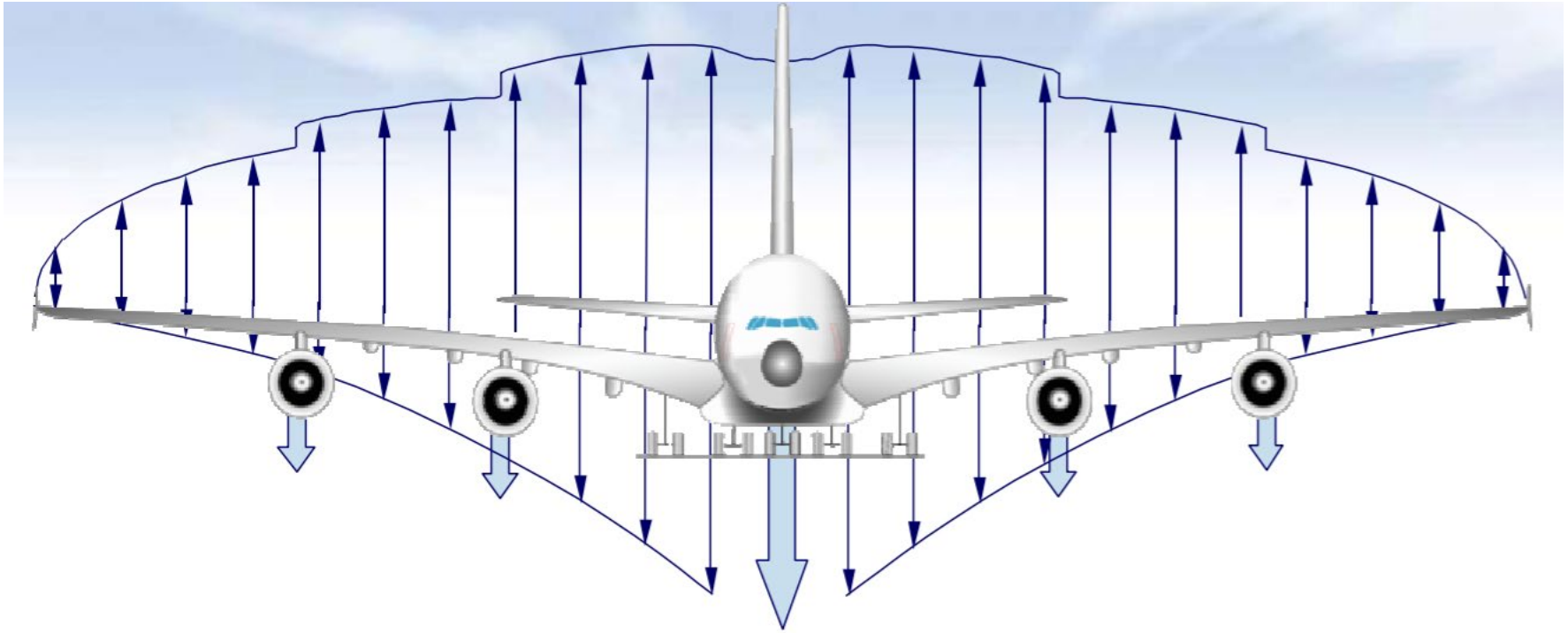
We test all linearly independent alternatives for the test function by setting one coefficient $\delta\hat{u}_j^{(n)} = 1$ and the rest equal to 0 each time. For each test (j,n), we get an equation:

$$\sum_i \sum_m K_{(j,n)(i,m)} \hat{u}_i^{(m)} = 0; \qquad K_{(j,n)(i,m)} = \sum_e K^{(e)}_{(j,l)(i,k)}$$

Only elements e such that element nodes $(e,l) \equiv n$ and $(e,k) \equiv m$

Motivation

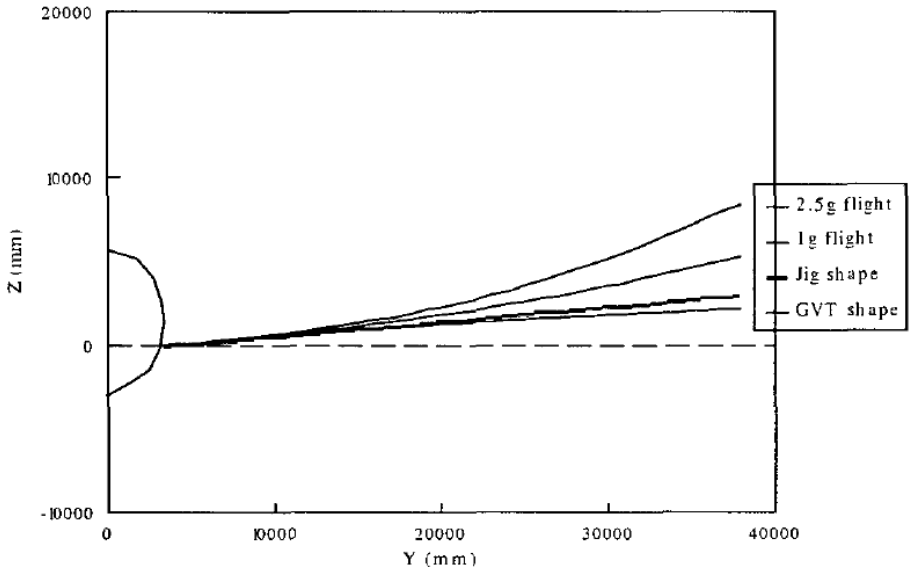
Example external loads applied on a wing



Motivation







Classical infinitesimal theory of elasticity (hypothesis)

1) $\|\boldsymbol{u}(x,t)\| \ll L_{\text{crit}}$ (small displacements, 0.1%)

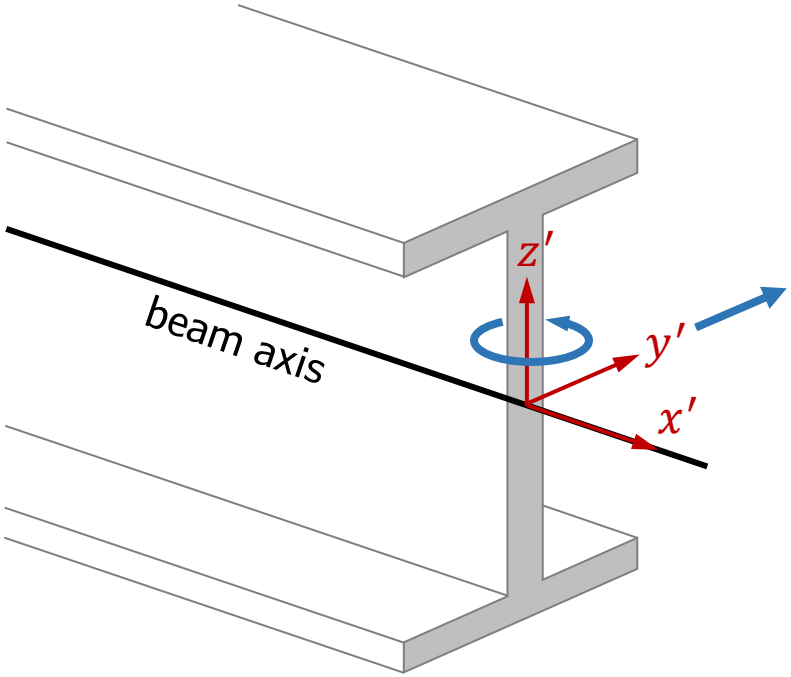
2) $\left\|\frac{\partial u_i}{\partial x_j}\right\| \ll 1$ (small strains, 0.1%)

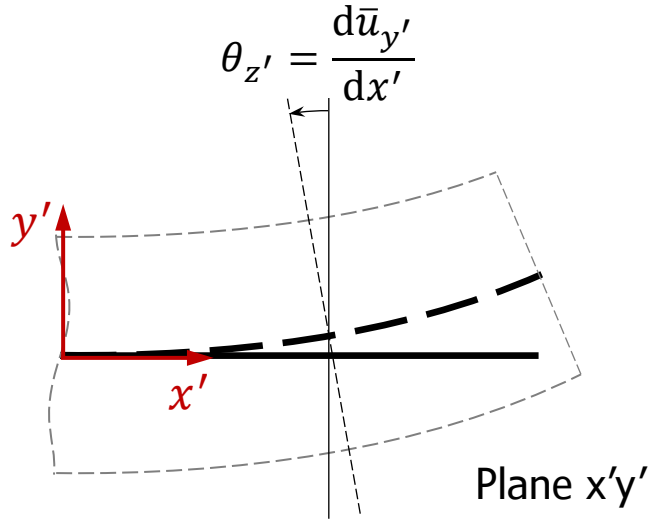
However, at in-flight 2.5g the tip deflection is 15% of the wingspan.

Beams theory

Hypothesis for *Euler-Bernoulli theory*:

- For ***shear loads*** applied on y' and/or ***bending moments*** about z':
  1) Deflection $u_{y'}$ of the points on a cross-section are small and equal to the deflection of the beam axis.
  2) Lateral displacement $u_{z'}$ is zero.
  3) Cross-sections to the beam axis remain plane.
  4) Cross-sections to the beam axis remain orthogonal to the beam axis after deformation.
- Equivalently, for ***shear loads*** applied on z' and/or ***bending moments*** about y':
  1) Deflection $u_{z'}$ of the points on a cross-section are small and equal to the deflection of the beam axis.
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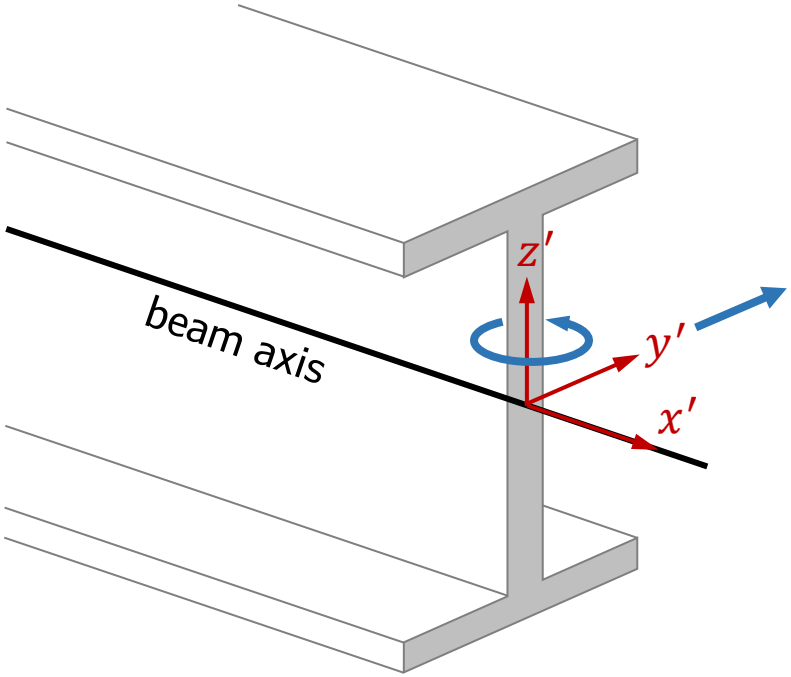


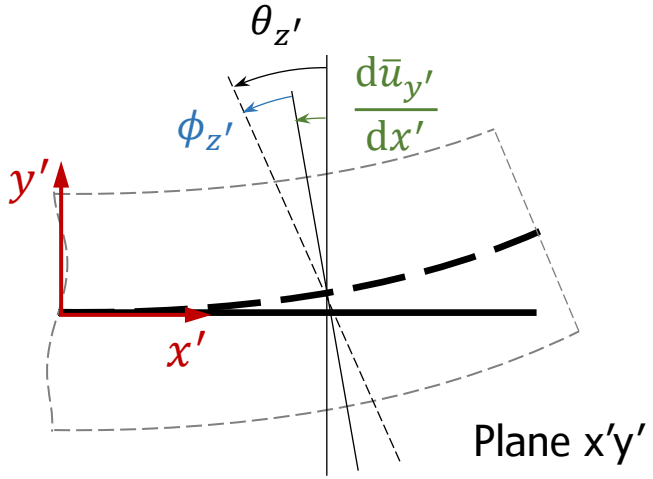


Beams theory

Hypothesis for *Timoshenko theory*:

- For ***shear loads*** applied on y' and/or ***bending moments*** about z':
  1) Deflection $u_{y'}$ of the points on a cross-section are small and equal to the deflection of the beam axis.
  2) Lateral displacement $u_{z'}$ is zero.
  3) Cross-sections to the beam axis remain plane.
  ~~4) Cross-sections to the beam axis remain orthogonal to the beam axis after deformation.~~
- Equivalently, for ***shear loads*** applied on z' and/or ***bending moments*** about y':
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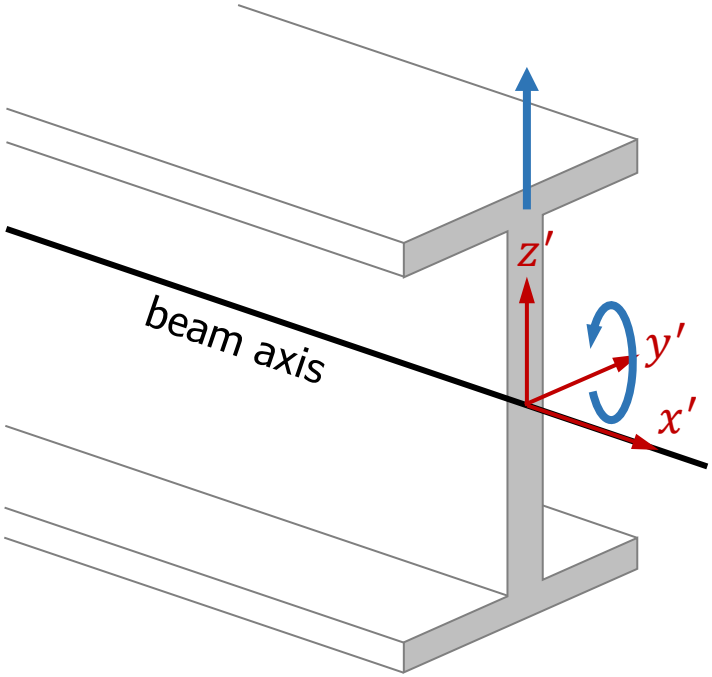


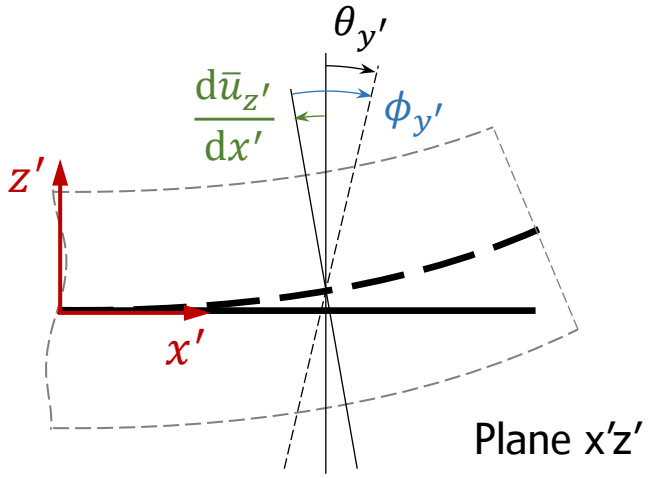
Beams theory

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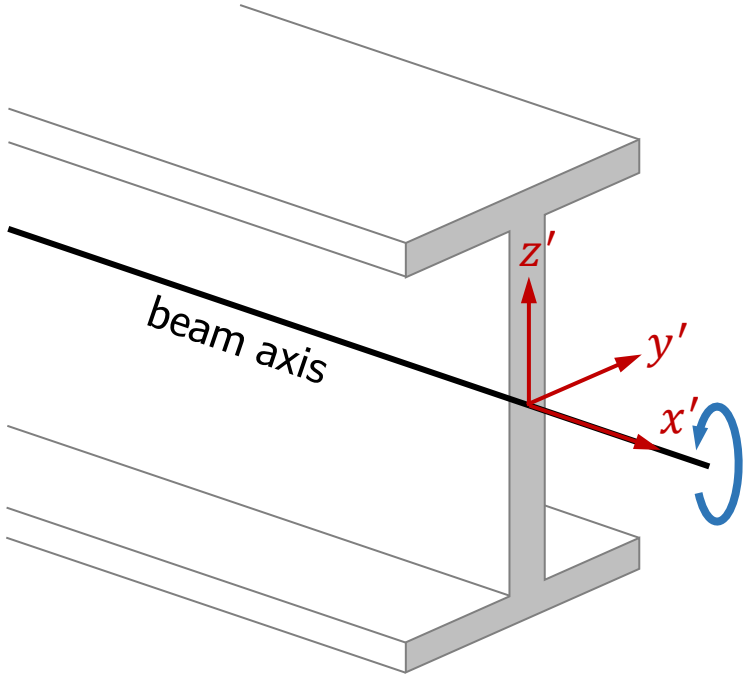


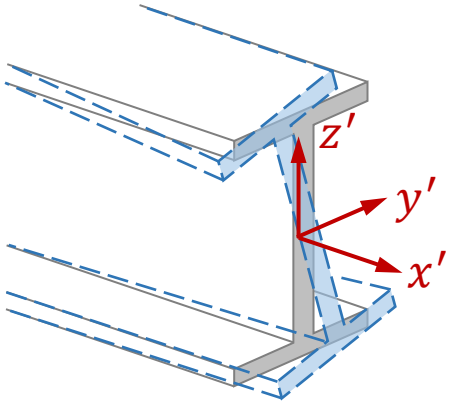
Beams theory

Torsion:

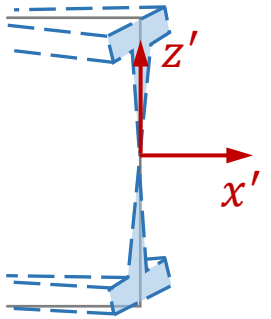
- For beams with circular cross-section subjected to a **torsion moment** about x' (*Saint-Venant theory*):
  1) The motion of each cross-section is a rigid body rotation about x'.
  2) All points in the cross-section remain plane (there are no displacements in the x' direction).
- For beams with non-circular cross-sections, hypothesis (2) is not satisfied, and a *warping displacement* on x' arises.

In practice, Saint-Venant theory is considered along with a torsion constant that generates an equivalence with the total deformation energy obtained when the warping displacement is accounted for.





Rigid-body rotation



Warping

Beams theory

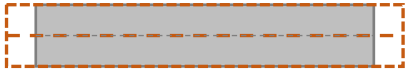
Local displacements:

$$u_{x'}(\boldsymbol{x}') = \bar{u}_{x'}(x') + z'\theta_{y'}(x') - y'\theta_{z'}(x')$$
$$u_{y'}(\boldsymbol{x}') = \bar{u}_{y'}(x') - z'\theta_{x'}(x')$$
$$u_{z'}(\boldsymbol{x}') = \bar{u}_{z'}(x') + y'\theta_{x'}(x')$$

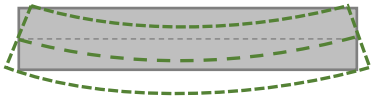
Local strain:

$$\varepsilon_{x'x'} = \frac{\partial u_{x'}}{\partial x'} = \frac{\mathrm{d}\bar{u}_{x'}}{\mathrm{d}x'} + z'\frac{\mathrm{d}\theta_{y'}}{\mathrm{d}x'} - y'\frac{\mathrm{d}\theta_{z'}}{\mathrm{d}x'}$$
$$\varepsilon_{y'y'} = \frac{\partial u_{y'}}{\partial y'} = 0$$
$$\varepsilon_{z'z'} = \frac{\partial u_{z'}}{\partial z'} = 0$$
$$\gamma_{x'y'} = \frac{\partial u_{y'}}{\partial x'} + \frac{\partial u_{x'}}{\partial y'} = \frac{\mathrm{d}\bar{u}_{y'}}{\mathrm{d}x'} - \theta_{z'} - z'\frac{\mathrm{d}\theta_{x'}}{\mathrm{d}x'} = -\phi_{z'} - z'\frac{\mathrm{d}\theta_{x'}}{\mathrm{d}x'}$$
$$\gamma_{x'z'} = \frac{\partial u_{z'}}{\partial x'} + \frac{\partial u_{x'}}{\partial z'} = \frac{\mathrm{d}\bar{u}_{z'}}{\mathrm{d}x'} + \theta_{y'} + y'\frac{\mathrm{d}\theta_{x'}}{\mathrm{d}x'} = \phi_{y'} + y'\frac{\mathrm{d}\theta_{x'}}{\mathrm{d}x'}$$
$$\gamma_{y'z'} = \frac{\partial u_{z'}}{\partial y'} + \frac{\partial u_{y'}}{\partial z'} = 0$$

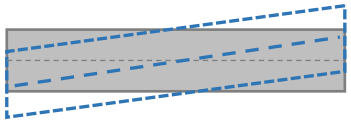
Axial



Bending



Shear



Torsion



Beams theory

Local displacements (vector form):

$$\{\boldsymbol{u}'\} = \begin{Bmatrix} u_{x'} \\ u_{y'} \\ u_{z'} \end{Bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & z' & -y' \\ 0 & 1 & 0 & -z' & 0 & 0 \\ 0 & 0 & 1 & y' & 0 & 0 \end{bmatrix}}_{[\mathbf{S}(y',z')]} \underbrace{\begin{Bmatrix} \bar{u}_{x'} \\ \bar{u}_{y'} \\ \bar{u}_{z'} \\ \theta_{x'} \\ \theta_{y'} \\ \theta_{z'} \end{Bmatrix}}_{\{\bar{\boldsymbol{u}}'(x')\}}$$

Local strain (vector form) decomposition:

$$\{\boldsymbol{\varepsilon}'\} = \begin{Bmatrix} \varepsilon_{x'x'} \\ \gamma_{x'y'} \\ \gamma_{x'z'} \end{Bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & z' & -y' \\ 0 & 1 & 0 & -z' & 0 & 0 \\ 0 & 0 & 1 & y' & 0 & 0 \end{bmatrix}}_{[\mathbf{S}(y',z')]} \underbrace{\begin{Bmatrix} \bar{u}_{x',x'} \\ \bar{u}_{y',x'} - \theta_{z'} \\ \bar{u}_{z',x'} + \theta_{y'} \\ \theta_{x',x'} \\ \theta_{y',x'} \\ \theta_{z',x'} \end{Bmatrix}}_{\{\bar{\boldsymbol{\varepsilon}}'(x')\}}$$

(*) Note on notation: $\dfrac{\partial a_i}{\partial x_j} = a_{i,j}$

Beams theory

E : Young's modulus
ν : Poisson's ratio
G : Shear modulus

Local stresses (vector form):

$$\{\boldsymbol{\sigma}'\} = \begin{Bmatrix} \sigma_{x'x'} \\ \tau_{x'y'} \\ \tau_{x'z'} \end{Bmatrix} = \underbrace{\begin{bmatrix} E & 0 & 0 \\ 0 & G & 0 \\ 0 & 0 & G \end{bmatrix}}_{[\mathbf{C}']} \begin{Bmatrix} \varepsilon_{x'x'} \\ \gamma_{x'y'} \\ \gamma_{x'z'} \end{Bmatrix}$$

$$G = \frac{E}{2(1+\nu)}$$

Local internal forces:

$$\{\bar{\boldsymbol{f}}'_{\text{int}}\} = \begin{Bmatrix} F_{x'} \\ F_{y'} \\ F_{z'} \\ M_{x'} \\ M_{y'} \\ M_{z'} \end{Bmatrix} = \int_A \begin{Bmatrix} \sigma_{x'x'} \\ \tau_{x'y'} \\ \tau_{x'z'} \\ y'\tau_{x'z'} - z'\tau_{x'y'} \\ z'\sigma_{x'x'} \\ -y'\sigma_{x'x'} \end{Bmatrix} \mathrm{d}A = \int_A \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -z' & y' \\ z' & 0 & 0 \\ -y' & 0 & 0 \end{bmatrix}}_{[\mathbf{S}(y',z')]^{\mathrm{T}}} \begin{Bmatrix} \sigma_{x'x'} \\ \tau_{x'y'} \\ \tau_{x'z'} \end{Bmatrix} \mathrm{d}A$$

Beams theory

Equilibrium equation (strong form):

$$\dot{\boldsymbol{p}}' = \boldsymbol{\nabla}_{x'} \cdot \boldsymbol{\sigma}' + \rho \boldsymbol{b}' + \boldsymbol{q}'$$

where $\boldsymbol{p}' = \rho \dot{\boldsymbol{u}}'$ is the linear momentum, ρ is the density (kg/m^3), $\boldsymbol{b}'$ are external body forces (N/kg) and $\boldsymbol{q}'$ are volume forces (N/m^3). The set of boundary conditions of the problem are:

- Prescribed displacement: $\boldsymbol{u}'|_{x' \in \Gamma_u} = \boldsymbol{u}'_{\mathrm{p}}$
- External traction force: $\boldsymbol{\sigma}' \cdot \hat{\boldsymbol{n}}|_{x' \in \Gamma_\sigma} = \boldsymbol{t}'$

Principle of virtual work (weak form):

$$\int_\Omega \{\delta \boldsymbol{\varepsilon}'\}^{\mathrm{T}} \{\boldsymbol{\sigma}'\} \mathrm{d}\Omega + \int_\Omega \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \rho \{\ddot{\boldsymbol{u}}'\} \mathrm{d}\Omega = \int_\Omega \{\delta \boldsymbol{u}'\}^{\mathrm{T}} (\rho \{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) \mathrm{d}\Omega + \int_{\Gamma_\sigma} \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \{\boldsymbol{t}'\} \mathrm{d}\Gamma$$

For all $\delta \boldsymbol{u}'|_{x' \in \Gamma_u} = \boldsymbol{0}$, and such that $\{\delta \boldsymbol{\varepsilon}'\}^{\mathrm{T}} = \left\{\delta u_{x',x'},\ \delta u_{x',y'} + \delta u_{y',x'},\ \delta u_{x',z'} + \delta u_{z',x'}\right\}$.

Assuming:

$$\{\delta \boldsymbol{u}'\} = [\mathbf{S}(y', z')]\{\delta \overline{\boldsymbol{u}}'(x')\} \quad \text{and} \quad \{\delta \boldsymbol{\varepsilon}'\} = [\mathbf{S}(y', z')]\{\delta \overline{\boldsymbol{\varepsilon}}'(x')\}$$

Beams theory

Focusing on each integral on the left-hand side:

- Stiffness term:

$$\int_\Omega \{\delta\boldsymbol{\varepsilon}'\}^{\mathrm{T}}\{\boldsymbol{\sigma}'\}\mathrm{d}\Omega = \int_\ell \{\delta\bar{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}} \underbrace{\left(\int_A [\mathbf{S}]^{\mathrm{T}}[\mathbf{C}'][\mathbf{S}]\mathrm{d}A\right)}_{[\bar{\mathbf{C}}']} \{\bar{\boldsymbol{\varepsilon}}'\}\mathrm{d}\ell$$

$$[\bar{\mathbf{C}}'] = \int_A [\mathbf{S}]^{\mathrm{T}}[\mathbf{C}'][\mathbf{S}]\mathrm{d}A = \begin{bmatrix} \overline{EA} & 0 & 0 & 0 & k_z^{\mathrm{ab}} & k_y^{\mathrm{ab}} \\ 0 & \overline{GA}_{y'} & 0 & k_z^{\mathrm{st}} & 0 & 0 \\ 0 & 0 & \overline{GA}_{z'} & k_y^{\mathrm{st}} & 0 & 0 \\ 0 & k_z^{\mathrm{st}} & k_y^{\mathrm{st}} & \overline{GJ}_{x'} & 0 & 0 \\ k_z^{\mathrm{ab}} & 0 & 0 & 0 & \overline{EI}_{y'} & \overline{EI}_{y'z'} \\ k_y^{\mathrm{ab}} & 0 & 0 & 0 & \overline{EI}_{y'z'} & \overline{EI}_{z'} \end{bmatrix}$$

$$\overline{EA} = \int_A E\mathrm{d}A\,;\ \ \overline{GA}_{y'} = k_y\int_A G\mathrm{d}A\,;\ \ \overline{GA}_{z'} = k_z\int_A G\mathrm{d}A\,;\ \ \overline{GJ}_{x'} = k_{\mathrm{t}}\int_A G\left(y'^2 + z'^2\right)\mathrm{d}A$$

$$\overline{EI}_{y'} = \int_A Ez'^2\mathrm{d}A\,;\ \ \overline{EI}_{z'} = \int_A Ey'^2\mathrm{d}A\,;\ \ \overline{EI}_{y'z'} = \int_A Ey'z'\mathrm{d}A$$

The coupling terms can be made 0 with an appropriate choice of the local reference frame!!

Beams theory

- Inertial term:

$$\int_\Omega \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \rho \{\ddot{\boldsymbol{u}}'\} \mathrm{d}\Omega = \int_\ell \{\delta \overline{\boldsymbol{u}}'\}^{\mathrm{T}} \underbrace{\left(\int_A [\mathbf{S}]^{\mathrm{T}} \rho [\mathbf{S}] \mathrm{d}A \right)}_{[\overline{\boldsymbol{\rho}}']} \{\ddot{\overline{\boldsymbol{u}}}'\} \mathrm{d}\ell$$

$$[\overline{\boldsymbol{\rho}}'] = \int_A [\mathbf{S}]^{\mathrm{T}} \rho [\mathbf{S}] \mathrm{d}A = \bar{\rho} A \begin{bmatrix} 1 & 0 & 0 & 0 & z'_{\mathrm{cm}} & -y'_{\mathrm{cm}} \\ 0 & 1 & 0 & -z'_{\mathrm{cm}} & 0 & 0 \\ 0 & 0 & 1 & y'_{\mathrm{cm}} & 0 & 0 \\ 0 & -z'_{\mathrm{cm}} & y'_{\mathrm{cm}} & r^2_{x'} & 0 & 0 \\ z'_{\mathrm{cm}} & 0 & 0 & 0 & r^2_{y'} & r^2_{y'z'} \\ -y'_{\mathrm{cm}} & 0 & 0 & 0 & r^2_{y'z'} & r^2_{z'} \end{bmatrix}$$

$$A = \int_A \mathrm{d}A\,; \quad \bar{\rho} = \frac{1}{A} \int_A \rho \mathrm{d}A\,; \quad y'_{\mathrm{cm}} = \frac{1}{\bar{\rho} A} \int_A \rho y' \mathrm{d}A\,; \quad z'_{\mathrm{cm}} = \frac{1}{\bar{\rho} A} \int_A \rho z' \mathrm{d}A$$

$$r^2_{x'} = \frac{1}{\bar{\rho} A} \int_A \rho \left(y'^2 + z'^2 \right) \mathrm{d}A\,; \quad r^2_{y'} = \frac{1}{\bar{\rho} A} \int_A \rho z'^2 \mathrm{d}A\,; \quad r^2_{z'} = \frac{1}{\bar{\rho} A} \int_A \rho y'^2 \mathrm{d}A\,; \quad r^2_{y'z'} = \frac{1}{\bar{\rho} A} \int_A \rho y' z' \mathrm{d}A$$

Remark: If the local reference frame is translated to the centre of mass, $y'_{\mathrm{cm}} = z'_{\mathrm{cm}} = 0$.

Beams theory

For beams with uniform (isotropic) cross-section properties that contain two axes of symmetry, matrices $[\bar{\mathbf{C}}']$ and $[\overline{\boldsymbol{\rho}}']$ simplify to:

$$[\bar{\mathbf{C}}'] = \begin{bmatrix} EA & 0 & 0 & 0 & 0 & 0 \\ 0 & k_y GA & 0 & 0 & 0 & 0 \\ 0 & 0 & k_z GA & 0 & 0 & 0 \\ 0 & 0 & 0 & k_t GJ & 0 & 0 \\ 0 & 0 & 0 & 0 & EI_{y'} & 0 \\ 0 & 0 & 0 & 0 & 0 & EI_{z'} \end{bmatrix} \quad [\overline{\boldsymbol{\rho}}'] = \rho \begin{bmatrix} A & 0 & 0 & 0 & 0 & 0 \\ 0 & A & 0 & 0 & 0 & 0 \\ 0 & 0 & A & 0 & 0 & 0 \\ 0 & 0 & 0 & J & 0 & 0 \\ 0 & 0 & 0 & 0 & I_{y'} & 0 \\ 0 & 0 & 0 & 0 & 0 & I_{z'} \end{bmatrix}$$

with:

- E : Young's modulus (N/m^2)
- G : Shear modulus (N/m^2) (Alternatively, if the Poisson's ratio ν is given, $G = E/2(1+\nu)$)
- ρ : Density (kg/m^3)
- A : Cross-section area (m^2)
- $I_{y'}$, $I_{z'}$: Area inertias of the section (m^2) ($I_{y'} = \int_A z'^2 \mathrm{d}A$ and $I_{z'} = \int_A y'^2 \mathrm{d}A$)
- J : Polar inertia of the section (m^2) ($J = \int_A (y'^2 + z'^2)\mathrm{d}A = I_{y'} + I_{z'}$)
- k_y, k_z, k_t : Shear and torsion correction constants

Remark: In this case, the neutral, shear and mass centres coincide with the intersection of the symmetry axes, where the origin of the local reference frame is located.

Beams theory

Now, focusing on each integral on the right-hand side:

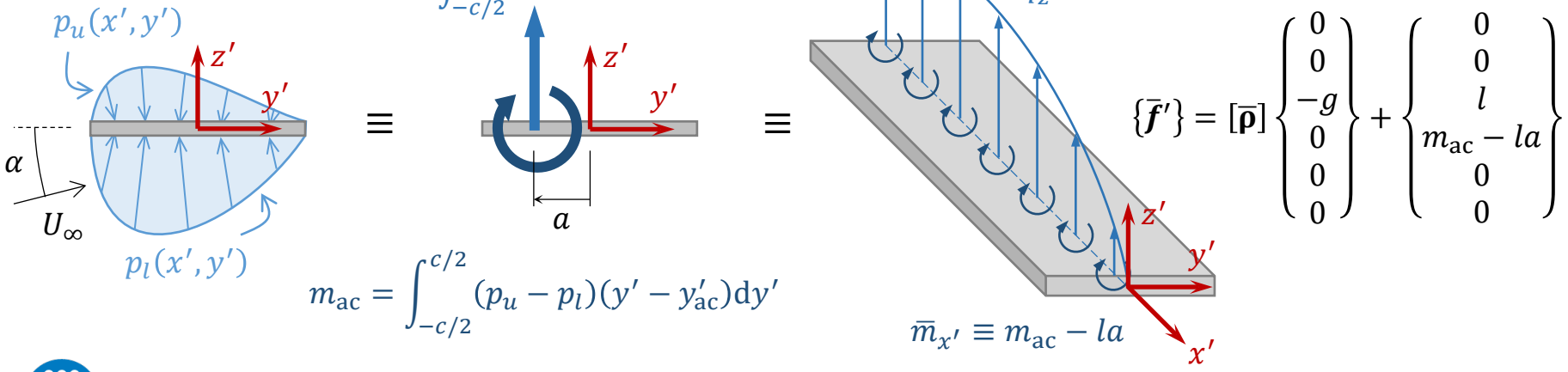
- External body/volume forces (assuming $\boldsymbol{b}'$ is uniform over the cross section):

$$\int_{\Omega} \{\delta \boldsymbol{u}'\}^{\mathrm{T}} (\rho\{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) \mathrm{d}\Omega = \int_{\ell} \{\delta \bar{\boldsymbol{u}}'\}^{\mathrm{T}} \left(\int_{A} [\mathbf{S}]^{\mathrm{T}} (\rho\{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) \mathrm{d}A \right) \mathrm{d}\ell$$

Distributed loads over beam (N/m)

$$\{\bar{\boldsymbol{f}}'\} = \int_{A} [\mathbf{S}]^{\mathrm{T}} (\rho\{\boldsymbol{b}'\} + \{\boldsymbol{q}'\}) \mathrm{d}A = [\bar{\boldsymbol{\rho}}'] \begin{Bmatrix} \bar{b}_{x'} \\ \bar{b}_{y'} \\ \bar{b}_{z'} \\ 0 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} \bar{q}_{x'} \\ \bar{q}_{y'} \\ \bar{q}_{z'} \\ \bar{m}_{x'} \\ \bar{m}_{y'} \\ \bar{m}_{z'} \end{Bmatrix} = [\bar{\boldsymbol{\rho}}'] \{\bar{\boldsymbol{b}}'\} + \{\bar{\boldsymbol{q}}'\}$$

Example:

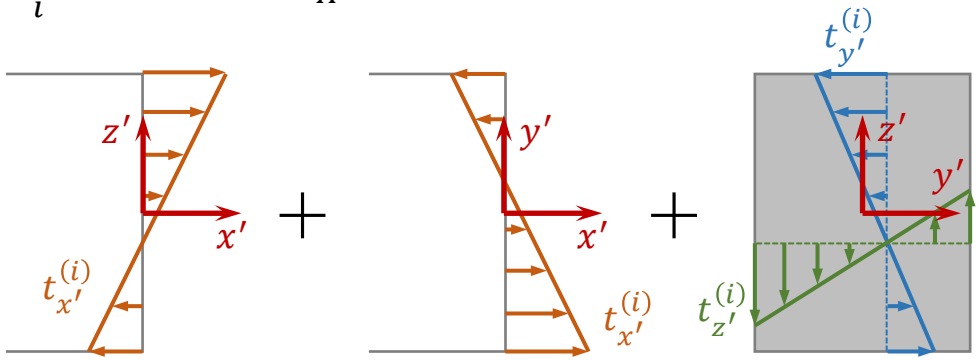


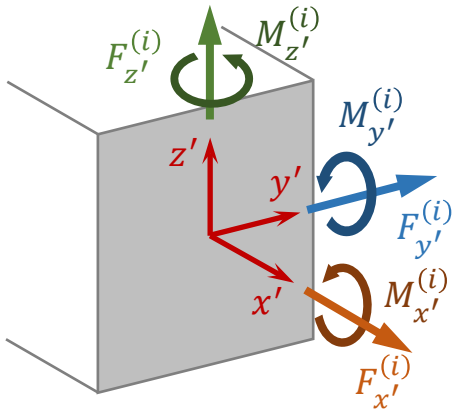
Beams theory

- External surface forces applied over specific cross-sections $x' = \{\ldots \quad x^{(i)} \quad \ldots\}$:

$$\int_{\Gamma_\sigma} \{\delta \boldsymbol{u}'\}^{\mathrm{T}} \{\boldsymbol{t}'\} \mathrm{d}\Gamma = \sum_i \left\{\delta \overline{\boldsymbol{u}}'\left(x^{(i)}\right)\right\}^{\mathrm{T}} \int_{A^{(i)}} [\mathbf{S}]^{\mathrm{T}} \left\{\boldsymbol{t}'^{(i)}\right\} \mathrm{d}A$$

$$\left\{\widehat{\boldsymbol{F}}^{(i)}\right\} = \int_{A^{(i)}} [\mathbf{S}]^{\mathrm{T}} \left\{\boldsymbol{t}'^{(i)}\right\} \mathrm{d}A = \begin{Bmatrix} F_{x'}^{(i)} \\ F_{y'}^{(i)} \\ F_{z'}^{(i)} \\ M_{x'}^{(i)} \\ M_{y'}^{(i)} \\ M_{z'}^{(i)} \end{Bmatrix}$$





$$F_{x'}^{(i)} = \int_{A^{(i)}} t_{x'}^{(i)} \mathrm{d}A$$

$$F_{y'}^{(i)} = \int_{A^{(i)}} t_{y'}^{(i)} \mathrm{d}A$$

$$F_{z'}^{(i)} = \int_{A^{(i)}} t_{z'}^{(i)} \mathrm{d}A$$

$$M_{x'}^{(i)} = \int_{A^{(i)}} \left(t_{z'}^{(i)} y' - t_{y'}^{(i)} z'\right) \mathrm{d}A$$

$$M_{y'}^{(i)} = \int_{A^{(i)}} t_{x'}^{(i)} z' \mathrm{d}A$$

$$M_{z'}^{(i)} = -\int_{A^{(i)}} t_{x'}^{(i)} y' \mathrm{d}A$$

Beams theory

The resulting system yields:

$$\int_\ell \{\delta\bar{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}}[\bar{\mathbf{C}}']\{\bar{\boldsymbol{\varepsilon}}'\}\mathrm{d}\ell + \int_\ell \{\delta\bar{\boldsymbol{u}}'\}^{\mathrm{T}}[\bar{\boldsymbol{\rho}}']\{\ddot{\bar{\boldsymbol{u}}}'\}\mathrm{d}\ell = \int_\ell \{\delta\bar{\boldsymbol{u}}'\}^{\mathrm{T}}\{\bar{\boldsymbol{f}}'\}\mathrm{d}\ell + \{\delta\bar{\boldsymbol{u}}'(x^{(i)})\}^{\mathrm{T}}\{\hat{\boldsymbol{F}}^{(i)}\}$$

The stiffness term can be further decomposed into different components:

$$\int_\ell \{\delta\bar{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}}[\bar{\mathbf{C}}']\{\bar{\boldsymbol{\varepsilon}}'\}\mathrm{d}\ell = \int_\ell \{\delta\bar{\boldsymbol{\varepsilon}}'_{\mathrm{a}}\}^{\mathrm{T}}[\bar{\mathbf{C}}'_{\mathrm{a}}]\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{a}}\}\mathrm{d}\ell + \int_\ell \{\delta\bar{\boldsymbol{\varepsilon}}'_{\mathrm{b}}\}^{\mathrm{T}}[\bar{\mathbf{C}}'_{\mathrm{b}}]\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{b}}\}\mathrm{d}\ell + \int_\ell \{\delta\bar{\boldsymbol{\varepsilon}}'_{\mathrm{s}}\}^{\mathrm{T}}[\bar{\mathbf{C}}'_{\mathrm{s}}]\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{s}}\}\mathrm{d}\ell + \int_\ell \{\delta\bar{\boldsymbol{\varepsilon}}'_{\mathrm{t}}\}^{\mathrm{T}}[\bar{\mathbf{C}}'_{\mathrm{t}}]\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{t}}\}\mathrm{d}\ell$$

$$\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{a}}\} = \{\bar{u}_{x',x'}\}; \quad [\bar{\mathbf{C}}'_{\mathrm{a}}] = [EA]$$

$$\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{b}}\} = \begin{Bmatrix}\theta_{y',x'}\\ \theta_{z',x'}\end{Bmatrix}; \quad [\bar{\mathbf{C}}'_{\mathrm{b}}] = \begin{bmatrix}EI_{y'} & 0\\ 0 & EI_{y'}\end{bmatrix}$$

$$\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{s}}\} = \begin{Bmatrix}\bar{u}_{y',x'} - \theta_{z'}\\ \bar{u}_{z',x'} + \theta_{y'}\end{Bmatrix}; \quad [\bar{\mathbf{C}}'_{\mathrm{s}}] = \begin{bmatrix}k_y GA & 0\\ 0 & k_z GA\end{bmatrix}$$

$$\{\bar{\boldsymbol{\varepsilon}}'_{\mathrm{t}}\} = \{\theta_{x',x'}\}; \quad [\bar{\mathbf{C}}'_{\mathrm{t}}] = [k_{\mathrm{t}} GJ]$$

Finite element discretization

Discretized displacements of element (e):

$$\left\{\overline{\boldsymbol{u}}'^{(e)}\right\} = \sum_i \begin{bmatrix} N^{(e,i)} & 0 & 0 & 0 & 0 & 0 \\ 0 & N^{(e,i)} & 0 & 0 & 0 & 0 \\ 0 & 0 & N^{(e,i)} & 0 & 0 & 0 \\ 0 & 0 & 0 & N^{(e,i)} & 0 & 0 \\ 0 & 0 & 0 & 0 & N^{(e,i)} & 0 \\ 0 & 0 & 0 & 0 & 0 & N^{(e,i)} \end{bmatrix} \begin{Bmatrix} \hat{u}_{x'}^{(e,i)} \\ \hat{u}_{y'}^{(e,i)} \\ \hat{u}_{z'}^{(e,i)} \\ \hat{\theta}_{x'}^{(e,i)} \\ \hat{\theta}_{y'}^{(e,i)} \\ \hat{\theta}_{z'}^{(e,i)} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{N}^{(e,i)} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Discretized *axial* strains of element (e):

$$\left\{\overline{\boldsymbol{\varepsilon}}_{\mathrm{a}}'^{(e)}\right\} = \sum_i \begin{bmatrix} N_{,x'}^{(e,i)} & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \hat{u}_{x'}^{(e,i)} \\ \hat{u}_{y'}^{(e,i)} \\ \hat{u}_{z'}^{(e,i)} \\ \hat{\theta}_{x'}^{(e,i)} \\ \hat{\theta}_{y'}^{(e,i)} \\ \hat{\theta}_{z'}^{(e,i)} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{B}_{\mathrm{a}}'^{(e,i)} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Finite element discretization

Discretized *bending* strains of element (e):

$$\left\{\bar{\boldsymbol{\varepsilon}}'^{(e)}_{\mathrm{b}}\right\} = \sum_i \begin{bmatrix} 0 & 0 & 0 & 0 & N^{(e,i)}_{,x'} & 0 \\ 0 & 0 & 0 & 0 & 0 & N^{(e,i)}_{,x'} \end{bmatrix} \begin{Bmatrix} \hat{u}^{(e,i)}_{x'} \\ \hat{u}^{(e,i)}_{y'} \\ \hat{u}^{(e,i)}_{z'} \\ \hat{\theta}^{(e,i)}_{x'} \\ \hat{\theta}^{(e,i)}_{y'} \\ \hat{\theta}^{(e,i)}_{z'} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{B}'^{(e,i)}_{\mathrm{b}} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Discretized *shear* strains of element (e):

$$\left\{\bar{\boldsymbol{\varepsilon}}'^{(e)}_{\mathrm{s}}\right\} = \sum_i \begin{bmatrix} 0 & N^{(e,i)}_{,x'} & 0 & 0 & 0 & -N^{(e,i)} \\ 0 & 0 & N^{(e,i)}_{,x'} & 0 & N^{(e,i)} & 0 \end{bmatrix} \begin{Bmatrix} \hat{u}^{(e,i)}_{x'} \\ \hat{u}^{(e,i)}_{y'} \\ \hat{u}^{(e,i)}_{z'} \\ \hat{\theta}^{(e,i)}_{x'} \\ \hat{\theta}^{(e,i)}_{y'} \\ \hat{\theta}^{(e,i)}_{z'} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{B}'^{(e,i)}_{\mathrm{s}} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Finite element discretization

Discretized *torsion* strains of element (e):

$$\left\{\bar{\boldsymbol{\varepsilon}}'^{(e)}_{\mathrm{t}}\right\} = \sum_i \begin{bmatrix} 0 & 0 & 0 & N^{(e,i)}_{,x'} & 0 & 0 \end{bmatrix} \begin{Bmatrix} \hat{u}^{(e,i)}_{x'} \\ \hat{u}^{(e,i)}_{y'} \\ \hat{u}^{(e,i)}_{z'} \\ \hat{\theta}^{(e,i)}_{x'} \\ \hat{\theta}^{(e,i)}_{y'} \\ \hat{\theta}^{(e,i)}_{z'} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{B}'^{(e,i)}_{\mathrm{t}} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{u}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Discretized *body force/distributed load* vector of element (e):

$$\left\{\bar{\boldsymbol{f}}'^{(e)}\right\} = \sum_i \begin{bmatrix} N^{(e,i)} & 0 & 0 & 0 & 0 & 0 \\ 0 & N^{(e,i)} & 0 & 0 & 0 & 0 \\ 0 & 0 & N^{(e,i)} & 0 & 0 & 0 \\ 0 & 0 & 0 & N^{(e,i)} & 0 & 0 \\ 0 & 0 & 0 & 0 & N^{(e,i)} & 0 \\ 0 & 0 & 0 & 0 & 0 & N^{(e,i)} \end{bmatrix} \begin{Bmatrix} \hat{f}^{(e,i)}_{x'} \\ \hat{f}^{(e,i)}_{y'} \\ \hat{f}^{(e,i)}_{z'} \\ \hat{m}^{(e,i)}_{x'} \\ \hat{m}^{(e,i)}_{y'} \\ \hat{m}^{(e,i)}_{z'} \end{Bmatrix} = \begin{bmatrix} \cdots & \mathbf{N}^{(e,i)} & \cdots \end{bmatrix} \begin{Bmatrix} \vdots \\ \hat{\boldsymbol{f}}'^{(e,i)} \\ \vdots \end{Bmatrix}$$

Finite element discretization

Rotation matrix (global to local coordinates) of element (e):

- For a given principal axis direction (either $\hat{\boldsymbol{j}}'^{(e)}$ or $\hat{\boldsymbol{k}}'^{(e)}$) in the global system (x, y, z):

  1) Obtain the beam orientation: $\hat{\boldsymbol{\imath}}'^{(e)} = \left(\hat{\boldsymbol{x}}^{(e,2)} - \hat{\boldsymbol{x}}^{(e,1)}\right) / \left\|\hat{\boldsymbol{x}}^{(e,2)} - \hat{\boldsymbol{x}}^{(e,1)}\right\|$

  2a) For a given $\hat{\boldsymbol{j}}'^{(e)}$: $\hat{\boldsymbol{k}}'^{(e)} = \hat{\boldsymbol{\imath}}'^{(e)} \times \hat{\boldsymbol{j}}'^{(e)}$

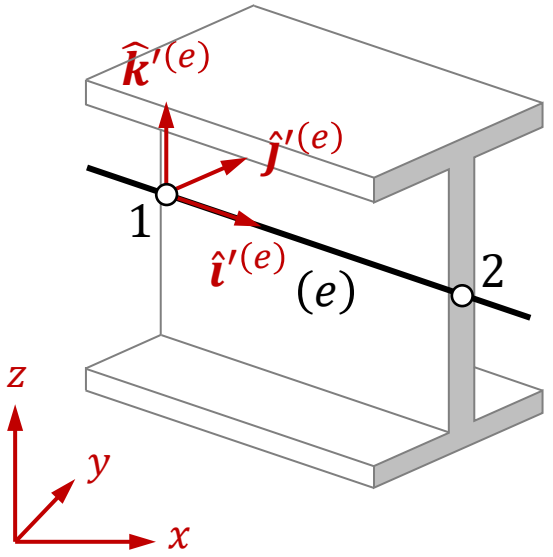
  2b) For a given $\hat{\boldsymbol{k}}'^{(e)}$: $\hat{\boldsymbol{j}}'^{(e)} = \hat{\boldsymbol{k}}'^{(e)} \times \hat{\boldsymbol{\imath}}'^{(e)}$

  Here, $\hat{\boldsymbol{x}}^{(e,i)}$ are the coordinates of node i of element (e) in the global system.

- The rotation matrix is then obtained as:

$$\left[\mathbf{R}^{(e)}\right] = \left[\hat{\boldsymbol{\imath}}'^{(e)} \quad \hat{\boldsymbol{j}}'^{(e)} \quad \hat{\boldsymbol{k}}'^{(e)}\right]^{\mathrm{T}} \rightarrow \quad \left\{\hat{\boldsymbol{a}}'\right\} = \left[\mathbf{R}^{(e)}\right]\left\{\hat{\boldsymbol{a}}\right\}$$

$$\underbrace{\begin{Bmatrix} \hat{u}_{x'}^{(e,i)} \\ \hat{u}_{y'}^{(e,i)} \\ \hat{u}_{z'}^{(e,i)} \\ \hat{\theta}_{x'}^{(e,i)} \\ \hat{\theta}_{y'}^{(e,i)} \\ \hat{\theta}_{z'}^{(e,i)} \end{Bmatrix}}_{\left\{\hat{\boldsymbol{u}}'^{(e,i)}\right\}} = \underbrace{\begin{bmatrix} i_x'^{(e)} & i_y'^{(e)} & i_z'^{(e)} & 0 & 0 & 0 \\ j_x'^{(e)} & j_y'^{(e)} & j_z'^{(e)} & 0 & 0 & 0 \\ k_x'^{(e)} & k_y'^{(e)} & k_z'^{(e)} & 0 & 0 & 0 \\ 0 & 0 & 0 & i_x'^{(e)} & i_y'^{(e)} & i_z'^{(e)} \\ 0 & 0 & 0 & j_x'^{(e)} & j_y'^{(e)} & j_z'^{(e)} \\ 0 & 0 & 0 & k_x'^{(e)} & k_y'^{(e)} & k_z'^{(e)} \end{bmatrix}}_{\left[\hat{\mathbf{R}}^{(e)}\right]} \underbrace{\begin{Bmatrix} \hat{u}_x^{(e,i)} \\ \hat{u}_y^{(e,i)} \\ \hat{u}_z^{(e,i)} \\ \hat{\theta}_x^{(e,i)} \\ \hat{\theta}_y^{(e,i)} \\ \hat{\theta}_z^{(e,i)} \end{Bmatrix}}_{\left\{\hat{\boldsymbol{u}}^{(e,i)}\right\}}$$



Finite element discretization

Discretized equation:

$$\sum_e \{\delta\hat{\boldsymbol{u}}^{(e)}\}^{\mathrm{T}} \left[[\mathbf{M}^{(e)}]\{\ddot{\hat{\boldsymbol{u}}}^{(e)}\} + \left(\left[\mathbf{K}_{\mathrm{a}}^{(e)}\right] + \left[\mathbf{K}_{\mathrm{b}}^{(e)}\right] + \left[\mathbf{K}_{\mathrm{s}}^{(e)}\right] + \left[\mathbf{K}_{\mathrm{m}}^{(e)}\right] \right) \{\hat{\boldsymbol{u}}^{(e)}\} - \{\hat{\boldsymbol{f}}^{(e)}\} \right] = 0$$

where:

$$[\mathbf{M}^{(e)}] = [\hat{\mathbf{R}}^{(e)}]^{\mathrm{T}} \int_{\ell^{(e)}} [\mathbf{N}^{(e)}]^{\mathrm{T}} [\bar{\boldsymbol{\rho}}'^{(e)}] [\mathbf{N}^{(e)}] \mathrm{d}\ell \, [\hat{\mathbf{R}}^{(e)}]$$

$$\left[\mathbf{K}_{\mathrm{a}}^{(e)}\right] = [\hat{\mathbf{R}}^{(e)}]^{\mathrm{T}} \int_{\ell^{(e)}} \left[\mathbf{B}_{\mathrm{a}}'^{(e)}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}_{\mathrm{a}}'^{(e)}\right] \left[\mathbf{B}_{\mathrm{a}}'^{(e)}\right] \mathrm{d}\ell \, [\hat{\mathbf{R}}^{(e)}]$$

$$\left[\mathbf{K}_{\mathrm{b}}^{(e)}\right] = [\hat{\mathbf{R}}^{(e)}]^{\mathrm{T}} \int_{\ell^{(e)}} \left[\mathbf{B}_{\mathrm{b}}'^{(e)}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}_{\mathrm{b}}'^{(e)}\right] \left[\mathbf{B}_{\mathrm{b}}'^{(e)}\right] \mathrm{d}\ell \, [\hat{\mathbf{R}}^{(e)}]$$

$$\left[\mathbf{K}_{\mathrm{s}}^{(e)}\right] = [\hat{\mathbf{R}}^{(e)}]^{\mathrm{T}} \int_{\ell^{(e)}} \left[\mathbf{B}_{\mathrm{s}}'^{(e)}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}_{\mathrm{s}}'^{(e)}\right] \left[\mathbf{B}_{\mathrm{s}}'^{(e)}\right] \mathrm{d}\ell \, [\hat{\mathbf{R}}^{(e)}]$$

$$\left[\mathbf{K}_{\mathrm{t}}^{(e)}\right] = [\hat{\mathbf{R}}^{(e)}]^{\mathrm{T}} \int_{\ell^{(e)}} \left[\mathbf{B}_{\mathrm{t}}'^{(e)}\right]^{\mathrm{T}} \left[\bar{\mathbf{C}}_{\mathrm{t}}'^{(e)}\right] \left[\mathbf{B}_{\mathrm{t}}'^{(e)}\right] \mathrm{d}\ell \, [\hat{\mathbf{R}}^{(e)}]$$

$$\{\hat{\boldsymbol{f}}^{(e)}\} = [\mathbf{M}^{(e)}]\{\hat{\boldsymbol{b}}^{(e)}\} + [\hat{\mathbf{R}}^{(e)}]^{\mathrm{T}} \int_{\ell^{(e)}} [\mathbf{N}^{(e)}]^{\mathrm{T}} [\mathbf{N}^{(e)}] \mathrm{d}\ell \, [\hat{\mathbf{R}}^{(e)}] \{\hat{\boldsymbol{q}}^{(e)}\}$$

Remember to add the effect of the external surface loads $\{\hat{\boldsymbol{F}}\}$ directly on the corresponding nodal degrees of freedom as point loads. Note: Beware to provide them in the global reference system!!

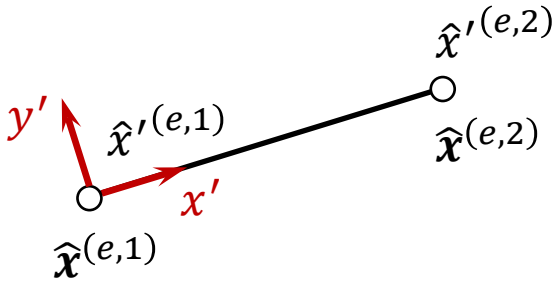
Global matrix system:

$$[\mathbf{M}]\{\ddot{\hat{\boldsymbol{u}}}\} + ([\mathbf{K}_{\mathrm{a}}] + [\mathbf{K}_{\mathrm{b}}] + [\mathbf{K}_{\mathrm{s}}] + [\mathbf{K}_{\mathrm{t}}])\{\hat{\boldsymbol{u}}\} = \{\hat{\boldsymbol{f}}\}$$

Linear elements integration

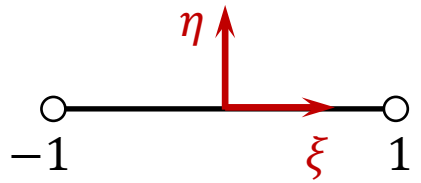
Shape functions in natural coordinates $\xi \in [-1, 1]$:

$$N^{(e,1)}(\xi) = \frac{1}{2}(1-\xi); \quad N^{(e,2)}(\xi) = \frac{1}{2}(1+\xi)$$



Shape functions derivatives in natural coordinates:

$$N^{(e,1)}_{,\xi} = -\frac{1}{2}; \quad N^{(e,2)}_{,\xi} = +\frac{1}{2}$$



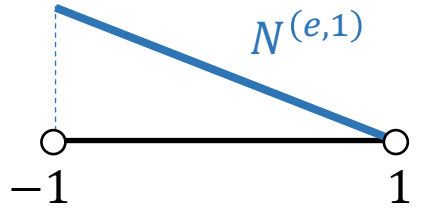
Jacobian:

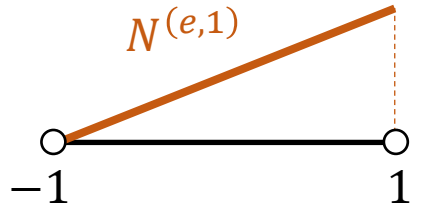
$$\mathcal{J}'^{(e)} = \frac{\partial x'}{\partial \xi} = \sum_i N^{(e,i)}_{,\xi} \underbrace{\{\hat{\boldsymbol{x}}^{(e,i)}\}^{\mathrm{T}}\{\hat{\boldsymbol{\imath}}'^{(e)}\}}_{\hat{x}'^{(e,i)}} = \frac{1}{2}\underbrace{(\hat{x}'^{(e,2)} - \hat{x}'^{(e,1)})}_{\ell^{(e)}}$$

Shape function derivatives in local coordinates:

$$N^{(e,1)}_{,x'} = \left(\mathcal{J}'^{(e)}\right)^{-1} N^{(e,1)}_{,\xi} = -\frac{1}{\ell^{(e)}}$$

$$N^{(e,2)}_{,x'} = \left(\mathcal{J}'^{(e)}\right)^{-1} N^{(e,2)}_{,\xi} = +\frac{1}{\ell^{(e)}}$$





Linear elements integration

Gauss quadrature integration:

$$\int_{\ell^{(e)}} f(x')\mathrm{d}x' = \sum_{k=1}^{N_\mathrm{G}} f(\xi_k)\mathcal{J}_k^{\prime(e)} w_k$$

where ξ_k and w_k represent the coordinate and weight of the k-th Gauss point, and N_G is the total number of Gauss points considered. Notice that $\mathcal{J}_k^{\prime(e)}$ is the Jacobian evaluated at ξ_k.

N_G	k	w_k	ξ_k
1	1	2	0
2	1	1	$-1/\sqrt{3}$
	2	1	$+1/\sqrt{3}$

For the mass matrix, 2 Gauss points provide the exact integration. For the shear component of the stiffness matrix, even though 2 Gauss points provide the exact integration, the element suffers from ***shear-locking*** for slender beams. A sub-integration with only 1 Gauss point avoids this problem at the expense of a small loss in accuracy. The remaining stiffness matrices do not require a Gauss quadrature to evaluate the integrals, since the integrands are constant over the element (in practice, 1 Gauss point may be used to compute the whole stiffness matrix).

Appendix A: Choice of reference frame

Choice of local reference frame:

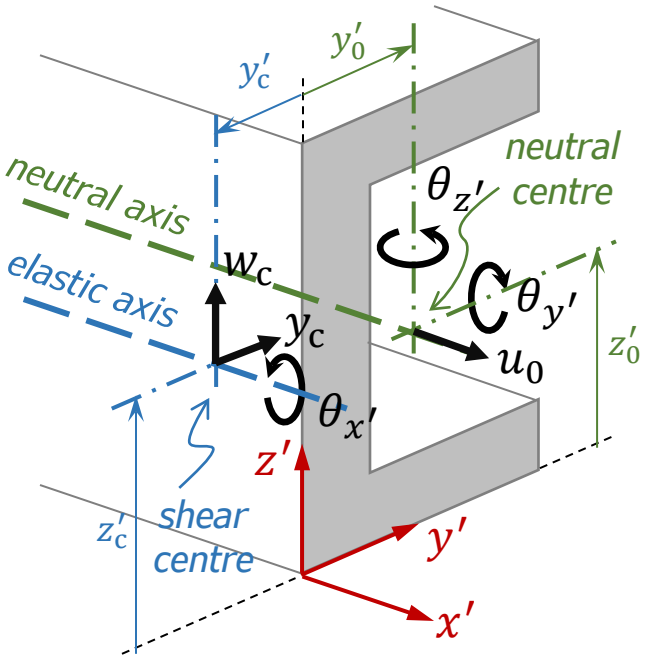
1) If axes of rotation y' and z' intersect at the *neutral centre* of the cross section, by definition, the axial and bending stresses and strains are uncoupled (i.e. $k_y^{\mathrm{ab}} = k_z^{\mathrm{ab}} = 0$). Furthermore, provided they are aligned with *principal directions* of the section, then also $\overline{EI}_{y'z'} = 0$.

2) If the axis of rotation x' intersects the cross section at the *shear centre*, by definition, the shear and torsion stresses and strains are uncoupled (i.e. $k_y^{\mathrm{st}} = k_z^{\mathrm{st}} = 0$).

Transformation:

$$\{\bar{\boldsymbol{u}}'\} = \begin{Bmatrix} \bar{u}_{x'} \\ \bar{u}_{y'} \\ \bar{u}_{z'} \\ \theta_{x'} \\ \theta_{y'} \\ \theta_{z'} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & -z_0' & y_0' \\ 0 & 1 & 0 & z_c' & 0 & 0 \\ 0 & 0 & 1 & -y_c' & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u_0 \\ v_c \\ w_c \\ \theta_{x'} \\ \theta_{y'} \\ \theta_{z'} \end{Bmatrix} = [\mathbf{T}]\{\tilde{\boldsymbol{u}}'\}$$

$$\{\tilde{\boldsymbol{u}}'\} = \begin{Bmatrix} u_0 \\ v_c \\ w_c \\ \theta_{x'} \\ \theta_{y'} \\ \theta_{z'} \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & z_0' & -y_0' \\ 0 & 1 & 0 & -z_c' & 0 & 0 \\ 0 & 0 & 1 & y_c' & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \bar{u}_{x'} \\ \bar{u}_{y'} \\ \bar{u}_{z'} \\ \theta_{x'} \\ \theta_{y'} \\ \theta_{z'} \end{Bmatrix} = [\mathbf{T}]^{-1}\{\bar{\boldsymbol{u}}'\}$$



Appendix A: Choice of reference frame

In this context, one can assume:

$$\int_{\Omega} \{\delta\boldsymbol{\varepsilon}'\}^{\mathrm{T}}\{\boldsymbol{\sigma}'\}d\Omega = \int_{\ell} \{\delta\tilde{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}}[\tilde{\mathbf{C}}]\{\tilde{\boldsymbol{\varepsilon}}'\}d\ell = \int_{\ell} \{\delta\bar{\boldsymbol{\varepsilon}}'\}^{\mathrm{T}}[\bar{\mathbf{C}}]\{\bar{\boldsymbol{\varepsilon}}'\}d\ell$$

$$[\tilde{\mathbf{C}}] = \begin{bmatrix} \overline{EA} & 0 & 0 & 0 & 0 & 0 \\ 0 & k_y\overline{GA} & 0 & 0 & 0 & 0 \\ 0 & 0 & k_z\overline{GA} & 0 & 0 & 0 \\ 0 & 0 & 0 & k_{\mathrm{t}}\overline{GJ}_{\mathrm{c}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \overline{EI}_{y_0'} & \overline{EI}_{y_0'z_0'} \\ 0 & 0 & 0 & 0 & \overline{EI}_{y_0'z_0'} & \overline{EI}_{z_0'} \end{bmatrix}$$

$$\overline{GJ}_{\mathrm{c}} = \int_A G\left((y'-y_c')^2 + (z'-z_c')^2\right)\mathrm{d}A$$

$$\overline{EI}_{y_0'} = \int_A E(z'-z_0')^2\mathrm{d}A\,;\ \overline{EI}_{z_0'} = \int_A E(y'-y_0')^2\mathrm{d}A\,;\ \overline{EI}_{y_0'z_0'} = \int_A E(y'-y_0')(z'-z_0')\mathrm{d}A$$

According to this, one gets:

$$[\bar{\mathbf{C}}] = ([\mathbf{T}]^{-1})^{\mathrm{T}}[\tilde{\mathbf{C}}]([\mathbf{T}]^{-1})$$

Appendix A: Choice of reference frame

In practice, the general constitutive tensor may be expressed as:

$$[\bar{\mathbf{C}}] = \begin{bmatrix} \overline{EA} & 0 & 0 & 0 & \overline{EA}z_0' & -\overline{EA}y_0' \\ 0 & k_y\overline{GA} & 0 & -\overline{GA}z_c' & 0 & 0 \\ 0 & 0 & k_z\overline{GA} & \overline{GA}y_c' & 0 & 0 \\ 0 & -\overline{GA}z_c' & \overline{GA}y_c' & k_t\overline{GJ}_c & 0 & 0 \\ \overline{EA}z_0' & 0 & 0 & 0 & \overline{EI}_{y_0'} & \overline{EI}_{y_0'z_0'} \\ -\overline{EA}y_0' & 0 & 0 & 0 & \overline{EI}_{y_0'z_0'} & \overline{EI}_{z_0'} \end{bmatrix}$$

Remarks:

- If the cross section has two axes of symmetry, the neutral centre coincides with the shear centre at the intersection between those axes.
- If the local reference frame is translated to the neutral centre of the cross section, the bending and axial components of the stiffness become uncoupled.
- If the local reference frame is translated to the shear centre of the cross section, the torsion and shear components of the stiffness become uncoupled.

